Question 1

Not complete

Marked out of 12.00

Write a complete Java code with main method to compute the following sum:

`m(i) = 1/3 + 2/5 + ... + i/(2i+1)`

Get i from the user and display the following table, for example k=2:

```
    i    m(i)
    1  0.3333
    2  0.7333
```

Hint: Put 10 character space for i and m(i).

Following is the outline of the program.

For example:

Input	Result
5	i m(i) 1 0.3333 2 0.7333 3 1.1619 4 1.6063 5 2.0609
11	i m(i) 1 0.3333 2 0.7333 3 1.1619 4 1.6063 5 2.0609 6 2.5224 7 2.9891 8 3.4597 9 3.9334 10 4.4096 11 4.8878

Answer: (penalty regime: 0 %)

Reset answer

```
1  //Here is the outline of the program:
2  public class Test {
3    public static void main(String[] args) {
4      // Fill in the code here
5
6
7
8
9
10
11
12
13
14
15
16
17   }
18
19   public static double m(int i) {
20     // Fill in the code here
21
22
23
24
25
26
27
28
29
30
31
```

```
32
33
34      }
35  }
36
```

Check

Question 2

Not complete

Marked out of 10.00

The Point2D class is defined as follows:

Point2D	
+Point2D(x: double, y: double)	Constructs a Point2D object with the specified x- and y-coordinates.
+distance(x: double, y: double): double	Returns the distance between this point and the specified point (x, y).
+distance(p: Point2D): double	Returns the distance between this point and the specified point p.
+getX(): double	Returns the x-coordinate from this point.
+getY(): double	Returns the y-coordinate from this point.
+midpoint(p: Point2D): Point2D	Returns the midpoint between this point and point p.
+distance(p1: Point2D, p2: Point2D): double	Returns the distance between p1 and p2.

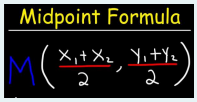
The class is partially implemented below. Implement distance(Point2D p1, Point2D p2) and midpoint(Point2D p).

You can use the following formula for calculating distance between two points:

Distance Between Two Points

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

You can use the following formula for calculating midpoint of two points:



For example:

Test	Result
Point2D p1 = new Point2D(4, 5); Point2D p2 = new Point2D(3.5, 5.6); System.out.printf("%.2f", Point2D.distance(p1, p2));	0.78
Point2D p3 = new Point2D(12.5, 5.8); Point2D p4 = new Point2D(4, 8.4); Point2D p5 = p3.midpoint(p4); System.out.printf("%.2f", Point2D.distance(p3, p5));	4.44

Answer: (penalty regime: 0 %)

Reset answer

```
1  class Point2D {
2    private double x;
3    private double y;
4
5    public Point2D(double x, double y) {
6      this.x = x;
7      this.y = y;
8    }
9
10   public double distance(double x, double y) {
11     return distance(new Point2D(x, y));
12   }
13
14   public double distance(Point2D p) {
15     return distance(this, p);
16   }
17
18   public static double distance(Point2D p1, Point2D p2) {
19   // Write code to implement it
20
21
22
23
24   }
25
26   public Point2D midpoint(Point2D p) {
27   // Write code to implement it
```

```
28
29
30
31
32    }
33
34    public double getX() {
35      return x;
36    }
37
38    public double getY() {
39      return y;
40    }
41  }
```

Check

Question 3

Not complete

Marked out of 20.00

Given a string and a number of lines k, print the string in twist form. In twist pattern, characters are printed out diagonally from top left to bottom right until reaching the kth line, then back up to top right, and so on.

✅ Hint 1: Think in terms of rows

Create an array or list of `k` rows (e.g., `StringBuilder[] rows`), each representing one line of the zigzag.

✅ Hint 2: Use a row index to track current position

Keep track of which row you're currently writing to with an index variable (e.g., `currentRow`).

✅ Hint 3: Use a direction flag

You'll need a flag to determine whether you're moving **down** or **up**. For example:

```
boolean goingDown = true;
```

When you reach the bottom (row `k-1`), switch direction to up. When you reach the top (row `0`), switch direction to down.

✅ Hint 4: Traverse the string character by character

Loop through each character in the string, and for each character:

- Add it to the correct row.
- Move the `currentRow` up or down depending on the current direction.

✅ Hint 5: To preserve visual structure, fill other rows with spaces

To keep the twist pattern shape visually aligned, you can add `' '` (spaces) to rows that are not the current one at each step.

✅ Hint 6: Print each row after processing

After placing all characters in rows, print each row line by line.

For example:

Input	Result
Çekoslovakyalılaştıramadıklarımızdanmışsınız 5	Ç a ş ı z ı e v k a t d k ı d s n k o y l ı a l m a ş ı o l a ı r m a ı n ı z s l a r m
Sozialversicherungsfachangestellte 6	S i a l o s c f c e l z r h s h t t i e e g a s e a v r n n e l u g
Incomprehensibilities 3	I m h i i s n o p e e s b l t e c r n i i

Answer: (penalty regime: 0 %)

```
1
```

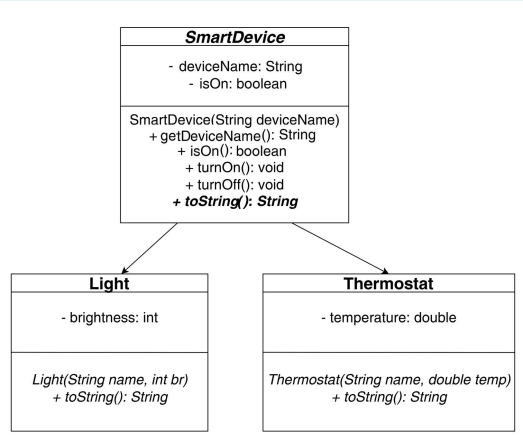
Check

Question 4

Not complete

Marked out of 30.00

Consider the following UML diagram.



Then develop a JAVA program to satisfy all the features given in it.

Read the below explanations.

SmartDevice (Abstract Class)

- **- deviceName: String** → The name of the device (e.g., "Living Room Light"). Set by constructor.
- **- isOn: boolean** → Represents whether the device is powered on or off.
- **+ getDeviceName(): String** → accessor method to get deviceName.
- **+ isOn(): boolean** → accessor method to get isOn.
- **smartDevice(String deviceName)** → Constructor.
- **+turnOn(): void** → Sets `isOn = true`. Can be called on any device.
- **+turnOff(): void** → Sets `isOn = false`. Common to all devices.
- **+toString(): String (abstract)** → Requires subclasses to provide a meaningful string representation (e.g., type + name + state).

Light (Subclass of SmartDevice)

- **- brightness: int** → Indicates brightness level (e.g., from 0 to 100%).
- **Light(String deviceName, int br)** → Constructor.
- **+toString(): String** → return example given below.

Thermostat (Subclass of SmartDevice)

- **- temperature: double** → Stores the current or target temperature in °C.
- **Thermostat(String deviceName, double temp)** → Constructor.
- **+toString(): String** → return example given below.

You also need to develop a test class that includes the main method.
Hint: You might need to clear newline after taking **int** or **double** values from the user. i.e. scanner.nextLine();

For example:

Input	Result
Living Room Lamp 80 Y Bedroom Thermostat 22.3 N	Enter light device name: Living Room Lamp Enter brightness (0-100): 80 Turn ON the light? (Y/N): Y Enter thermostat device name: Bedroom Thermostat Enter temperature (C): 22.3 Turn ON the thermostat? (Y/N): N SUMMARY: Light 'Living Room Lamp' is ON, Brightness: 80% Thermostat 'Bedroom Thermostat' is OFF, Temperature: 22.3C

Answer: (penalty regime: 0 %)

```
1
```

Check

Question 5

Not yet answered

Marked out of 28.00

Consider the following Java classes:

```java
// Interface: Insurable
interface Insurable {
    double calculateInsurance();
}

// Abstract class: Vehicle
abstract class Vehicle implements Insurable {
    private String brand;
    private int speed;
    private static int vehicleCount = 0;

    public Vehicle(String brand, int speed) {
        this.brand = brand;
        this.speed = speed;
        vehicleCount++;
    }

    public String getBrand() {
        return brand;
    }

    public void setBrand(String brand) {
        this.brand = brand;
    }

    public int getSpeed() {
        return speed;
    }

    public void setSpeed(int speed) {
        this.speed = speed;
    }

    public static int getVehicleCount() {
        return vehicleCount;
    }

    public abstract void start();

    public void stop() {
        System.out.println(brand + " stopped.");
    }
}
```

```java
// Subclass: Car
class Car extends Vehicle {
    private boolean isElectric;

    public Car(String brand, int speed, boolean isElectric) {
        super(brand, speed);
        this.isElectric = isElectric;
    }

    public boolean isElectric() {
        return isElectric;
    }

    public void setElectric(boolean electric) {
        isElectric = electric;
    }

    @Override
    public void start() {
        System.out.println(getBrand() + " car is starting...");
    }

    @Override
    public double calculateInsurance() {
        return getSpeed() * (isElectric ? 1.2 : 1.5);
    }
}

// Subclass: Motorcycle
class Motorcycle extends Vehicle {
    private boolean hasSidecar;

    public Motorcycle(String brand, int speed, boolean hasSidecar) {
        super(brand, speed);
        this.hasSidecar = hasSidecar;
    }

    public boolean hasSidecar() {
        return hasSidecar;
    }

    public void setSidecar(boolean hasSidecar) {
        this.hasSidecar = hasSidecar;
    }

    @Override
    public void start() {
        System.out.println(getBrand() + " motorcycle is starting...");
    }

    @Override
    public double calculateInsurance() {
        return getSpeed() * (hasSidecar ? 1.1 : 1.3);
    }
}
```

For each part below, write a single Java instruction that can be placed within the Test class with main to perform the indicated task.(You may assume that all objects have been assigned some value by the execution of the prior code.)

Hint: Put empty space after commas using Constructors.

Important Note: Note that if you misspell even one character, you will not get any points from the relevant field!

Write a statement to create a `Car` object called `electricCar` with `TOGG` brand electric car and speed `130`.

Car electricCar = new Car("TOGG", 130, true);

Write a statement to create a `Motorcycle` object named `sportBike` with brand `Kanuni`, speed `110`, and no sidecar.

Motorcycle sportBike = new MotorCycle("Kanuni", 110, false);

Write a statement by using a `Vehicle` reference to hold a `Car` object with brand `Audi`, speed `100`, and not electric. Name the variable `v1`.

Vehicle v1 = new Car("Audi", 100, false);

Write an instruction for printing the number of vehicles created so far on the screen and move the cursor to the next line:

System.out.println(Vehicle.getVehicleCount());

Write an instruction for increasing `bike1` speed by 10 (**Do not** print): bike1.setSpeed()(bike1.getSpeed() + 10);

What will be the insurance value for a new electric car with a speed of 120? Give integer value: 120*1,2=144

How do you calculate the insurance value of your `Motorcycle` object named `sportBike?` (**Do not** print).

```
sportBike.calculateInsurance();
```

Static olmadığı için verilen nesne adıyla çağırdık.
Static olsaydı class adıyla çağırırdık ve bize de nesne adı verilmezdi zaten.